



DUNGEON TRIAL

A Prototype Game Built with Unreal Engine 5

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What is Unreal Engine 5?

- Industry-standard game engine by Epic Games
- Known for high-quality visuals & real-time rendering
- Includes Nanite (virtualized geometry)
- Includes Lumen (dynamic global illumination)
- Hybrid workflow: C++ & Blueprints



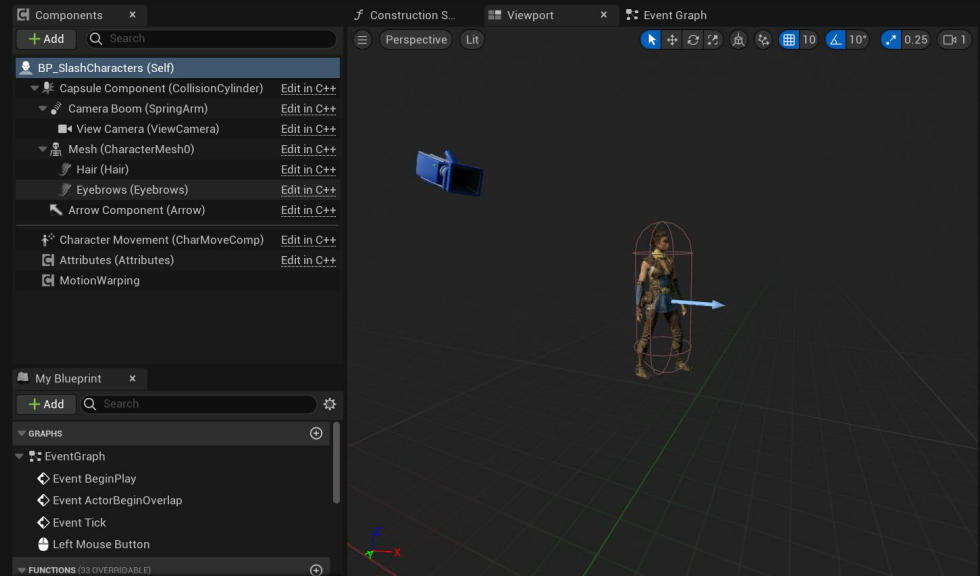
Why I Chose UE5

- Balance between visual quality and flexible coding
- Ideal for action-adventure/dungeon gameplay
- Provides insight into professional studio workflows
- Great documentation and community support



Project Goal

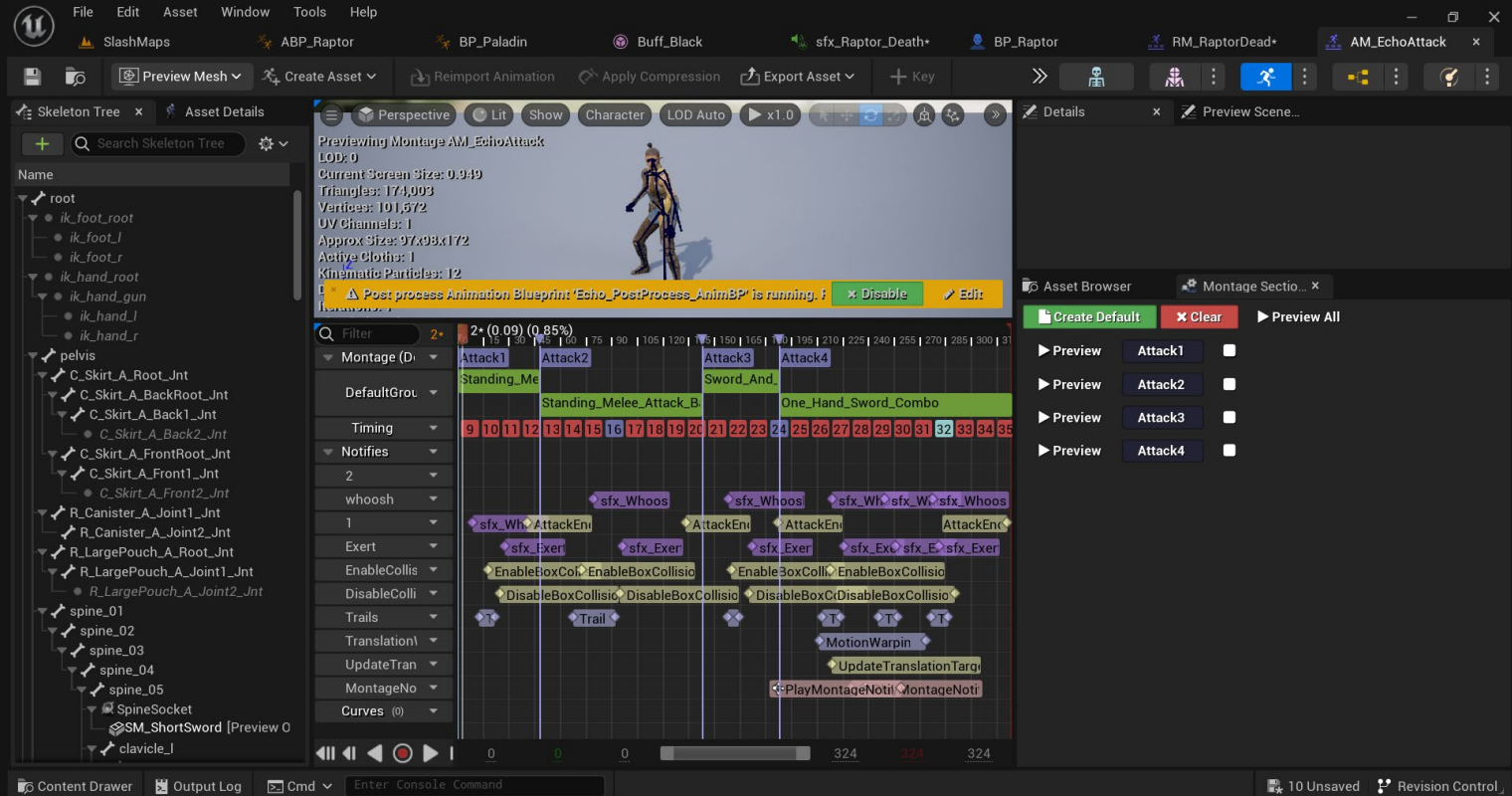
- Purpose: Learn core development concepts
- Focus on Level Design & Player Movement



Project Goal

- Understanding Combat Logic
- Learning Animation Workflows
- Prototyping ideas quickly before full production

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Game Summary

- A small dungeon-style action prototype
- Gameplay Loop: Explore -> Fight -> Progress
- Goal: Understand how UE5 systems work together
- Not a full game, but playable

Assets Used

- Character: UE5 Mannequin / Mixamo / Fab
- Environment: Dungeon assets (Marketplace)
- Animations: Idle, Run, Attack (Mixamo)
- Audio: Footsteps, Sword SFX
- VFX: Simple particles and fire

Why C++?

- Foundation of gameplay systems
- Better performance & deeper control
- Used for:
 - - Player Movement
 - - Attack Mechanics
 - - Enemy AI Logic

Why Blueprints?

- Visual scripting for rapid iteration
- Great for non-code systems
- Used for:
 - - Animations (Blendspaces)
 - - UI Elements
 - - Simple Triggers (Doors)

What I Learned

- The full workflow: Asset import to playable build
- Separating Systems: C++ (Logic) vs Blueprint (Visuals)
- Comfort with UE5 interface and Debugging tools

Conclusion

- Dungeon Trial strengthened my UE5 fundamentals
- Combining C++ and Blueprint is powerful
- This prototype sets the foundation for future projects